

SIGN LANGUAGE INTERPRETER

Ayushi
24WH1A0554

Lakshmi
24WH1A0501

Under the Guidance of
Dr. Aruna Rao
Head of Department
Dept of C.S.E
BVRIT Hyderabad College of Engineering for Women

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The Challenge

- Lack of accessible real-time sign language interpretation tools

Our Approach

- Computer vision based system
- Recognizes ASL gestures in real-time
- Converts gestures into text and speech

- Recognize 26 ASL alphabets (A–Z)
- Achieve 93%+ accuracy
- Maintain real-time processing (20 FPS)
- Run on standard devices
- Convert gestures to text and speech

- Real-time alphabet recognition
- Word builder using gesture buffering
- Text-to-speech output
- Live webcam with hand skeleton
- Web-based interface (Streamlit)

- ① Webcam Input
- ② Hand Detection (MediaPipe)
- ③ Feature Extraction (21 landmarks)
- ④ ML Classification (Random Forest)
- ⑤ Stability Check
- ⑥ Text + Speech Output

Core Technologies

- Python
- MediaPipe
- OpenCV
- Scikit-learn
- Streamlit
- pyttsx3 / Google TTS

- ASL Alphabet Dataset (87K images)
- WLASL Dataset (21K+ videos)
- Custom Dataset (6500+ samples)

- Algorithm: Random Forest
- 200 decision trees
- Input: 42 features
- Output: Alphabet + confidence

Performance:

- Training Accuracy: 96%
- Test Accuracy: 93–95%

- ① Capture frame
- ② Detect hand
- ③ Extract features
- ④ Predict gesture
- ⑤ Stability check
- ⑥ Update text buffer
- ⑦ Generate speech

- Live webcam feed
- Real-time text display
- Confidence score
- Word history
- Control buttons
- Audio output

Current

- Documentation completed
- Requirements gathered
- Data collection in progress

Upcoming

- Model training
- Real-time integration
- UI development

- Word-level recognition (LSTM)
- Mobile application
- Cloud deployment

- Trained ML model
- Web application
- Source code
- Demo video
- Documentation

Thank You